Homework 4

ASEN 5335 Aerospace Environment

Justin Le
October 18, 2026

Problem 1: Field Magnitudes

Using the WMM model from the NOAA calculator, we find the magnetic field in Boulder, CO to be,

Model Used:	WMM-2025						
Latitude:	40° 0' 59" N						
Longitude:	105° 17' 10" W						
Elevation:	5430.0 ft Mean Sea Level						
Date	**Declination (+ E | - W)**	**Inclination (+ D | - U)**	**Horizontal Intensity**	**North Comp (+ N | - S)**	**East Comp (+ E | - W)**	**Vertical Comp (+ D | - U)**	**Total Field**
2026-10-18	7° 40' 8"	66° 0' 55"	20,760.4 nT	20,574.7 nT	2,770.4 nT	46,662.2 nT	51,072.1 nT
Change/year	-0° 4' 50"/yr	-0° 2' 59"/yr	-8.9 nT/yr	-5.0 nT/yr	-30.2 nT/yr	-129.4 nT/yr	-121.9 nT/yr
Uncertainty	0° 22'	0° 12'	133 nT	137 nT	89 nT	141 nT	138 nT

Figure 1: Boulder Magnetic Field

We repeat with the same model for McMurdo Station Anarctica to find,

Model Used:	WMM-2025						
Latitude:	77° 50' 46" S						
Longitude:	166° 40' 5" E						
Elevation:	33.0 ft Mean Sea Level						
Date	**Declination (+ E | - W)**	**Inclination (+ D | - U)**	**Horizontal Intensity**	**North Comp (+ N | - S)**	**East Comp (+ E | - W)**	**Vertical Comp (+ D | - U)**	**Total Field**
2026-10-18	140° 5' 50"	-80° 5' 50"	10,647.5 nT	-8,168.0 nT	6,830.2 nT	-60,990.6 nT	61,913.0 nT
Change/year	-0° 12' 10"/yr	0° 2' 29"/yr	33.8 nT/yr	-1.7 nT/yr	50.6 nT/yr	66.8 nT/yr	-60.0 nT/yr
Uncertainty	0° 34'	0° 12'	133 nT	137 nT	89 nT	141 nT	138 nT

Figure 2: McMurdo Magnetic Field

TODO: Based on the inclination observed at McMurdo, this tells us that...

Problem 2: Dipole Field

Problem 3: Magnetopause

Problem 4: The Ring Current

Problem 5: Drift due to Gravity

Problem 6: Geomagnetic Indices

Problem 7: Geomagnetic Storm

Problem 8: Key Parameters in the Solar Wind Driving a Magnetic Storm

Problem 9: The Core

Problem 10: Magnetospheric Multiscale